

Low-Cost Initial Test (India, ≤ USD 10,000)

Can a plasticity-opening treatment (FGF17 + 7,8-DHF) combined with a structured learning environment raise performance across different cognitive challenges in rats? A feasibility study that decides whether a larger experiment is justified

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Document	Adapted initial-test protocol (India), derived from the validated five-round protocol v5
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Validation	Two rounds of three independent Claude Opus critics (design/statistics; pharmacology/delivery; enrichment/plasticity/logistics) plus a Claude Haiku reference and price check per round; one extra Opus review of the human-translation goal.
Status	Every reviewer verdict was "approve with changes"; the changes are applied below. Not yet approved by an ethics committee. Prices are estimates, not quotes.

Executive summary

Question. Does a plasticity-opening treatment — FGF17 (the factor identified in young cerebrospinal fluid, Iram 2022) plus 7,8-DHF (a putative TrkB/BDNF-pathway agonist) — combined with a structured, enriched learning environment (ELE) make healthy young rats score higher than their untreated peers across several different cognitive tasks, including one that was never trained (transfer)? Assumption to confirm: "crf" in the request was read as CSF.

What this is. A small, inexpensive first test (about 64 rats enrolled, ~10 completers per group). It is a feasibility, variance and direction study with pre-registered go/no-go rules. It is deliberately **not** powered to confirm an effect and cannot show "intelligence", "neuroplasticity" or any human benefit. If the gates are met, the larger experiment (Stage 1 in section 2, gates in section 8) is justified.

What was cut compared with the full protocol: aged animals, sex factor, the lifespan module, osmotic pumps (about USD 110–150 each; 48 pumps would have used more than half of the budget), R13, ANA-12, LTP, EM, Golgi-Cox, Luxol Fast Blue and the 40% attrition buffers. **Estimated cost:** about USD 9.2k base plus a reserve of about 0.8k, with documented cuts if prices run high (section 9).

Long-term goal (defined by the researcher, not tested here): to develop this combined approach toward cognitive gain in people with learning disabilities / altered neural connectivity and in older adults with Alzheimer's disease, by supporting the repair of neuronal connections. This is stated as a hypothesis and a roadmap in section 2, with the known risks.

1. Aim, hypotheses and what the data can claim

- **Primary aim (renamed "feasibility"):** estimate completion rate, variance and direction of an aggregate cognitive-performance index in a 2x2 of drug package x learning environment, and test the practical assays.
- **Exploratory hypothesis:** D+E (drug package + enrichment) scores higher than control peers (C) and higher than either component alone (D, E).
- **Transfer:** performance on a task that is categorically different from anything practised in the enrichment programme.
- **Allowed claims:** feasibility, attrition, variance of the index, direction and 95% interval of the D+E – C difference, assay performance, and descriptive marker correlates, for the drug package as a whole.
- **Not allowed:** "smarter", "general intelligence" or a "g factor" (an aggregate of 4 tasks at n≈10 is not a validated general factor); neuroplasticity (no LTP, spines or causal manipulation); single-agent claims (FGF17 vs 7,8-DHF are not separated); TrkB engagement unless the validated western gate passes; any human or disease claim; replication of Iram 2022 (that study rescued aged mice; this one asks about enhancement in healthy young adults, which is untested in the literature found).

2. Long-term goal and translational roadmap (later experiments, not this study)

Recommended wording (translational reviewer, adopted):

"Our long-term aim is to test whether combining a plasticity-promoting biological treatment (an FGF17-type factor plus a validated TrkB/BDNF-pathway agent) with structured enriched learning can produce measurable cognitive benefit in humans. We pursue two **separate** programmes with distinct biology, models, ages and ethics: (a) neurodevelopmental conditions with altered connectivity (learning disabilities), and (b) age-related cognitive decline and early Alzheimer's disease. We hypothesise that supporting synaptic and myelin plasticity is **one possible contributor** to such benefit; this is a hypothesis to be demonstrated with structural and functional evidence, not an assumption, and it is not addressed by the present study. Prior neurotrophin programmes in Alzheimer's have largely failed, chiefly on delivery, and both candidate agents face unresolved questions (human CNS delivery of FGF17; whether 7,8-DHF engages TrkB). Each stage proceeds only on pre-registered target-engagement, structural and safety gates."

Stage	Content	Gate to proceed
0 – this study	Healthy young adult rats; feasibility, variance, direction; engagement and structural markers.	Go/no-go rules, section 8.
1 – powered rat confirmation	Both sexes; full factorial separating FGF17, 7,8-DHF and enrichment; sustained infusion (pump); dose-response; durable vs on-drug effects; necessity tests (TrkB antagonist; FGFR3 knockdown).	Replicated effect on pre-registered primary; structural effect that co-varies with behaviour.
2 – disease models (separate tracks)	Aged rodents (Iram-type rescue); Alzheimer's models (5xFAD, APP/PS1, TgF344-AD) with the caveat that amyloid-model rescue has repeatedly failed to translate; neurodevelopmental models treated at developmentally appropriate ages — Fmr1 knockout (fragile X) and Down-syndrome models (Ts65Dn is being superseded by newer models) as separate tracks, not lumped as "learning disability".	Benefit in disease models with engagement and structural evidence.
3 – drug development	A clinically deliverable form (intranasal, intrathecal or engineered/receptor-selective agent — all unproven); a validated TrkB-pathway drug with human-relevant PK; two-species toxicology; FGF-class risks (mitogenesis/tumour promotion, angiogenesis); regulatory plan (CDSCO New Drugs and Clinical Trials Rules 2019, ICMR ethics, US/EU IND).	Delivery solved; clean toxicology.
4 – human studies	Phase 1 safety/PK; biomarkers (MRI myelin water fraction/DTI, EEG, standard cognitive tests); structured learning input as digital cognitive training; Phase 2 separately in older adults with MCI/early AD and in neurodevelopmental populations. Children and cognitively impaired adults: special DCGI scrutiny, assent plus proxy consent, independent data-safety monitoring.	Safety and biomarker signal.

What is not known (must be stated in any publication): no human trials of FGF17 or 7,8-DHF were found in this review (verify before citing); no validated way exists to deliver FGF17 to the human brain — a programme-ending risk; ICV delivery, the route used in this rat study, penetrates brain tissue poorly, which is the leading explanation for earlier NGF/BDNF failures in Alzheimer's; 7,8-DHF is reported not to activate TrkB reliably in cell assays (Boltaev 2017; Todd 2014; a 2021 screen); healthy-animal enhancement may not transfer to disease or humans; the real FGF-class safety concerns are mitogenesis and angiogenesis, not the FGF23/phosphate effect (a different, endocrine FGF).

What this rat study must measure to earn Stage 1: FGF17 engagement (pERK in the satellite), 7,8-DHF engagement (validated hippocampal western), a structural marker (BrdU+/Olig2+ cells, MBP) that co-varies with behaviour, an untrained-task

(transfer) effect, and separation of drug from enrichment (arms D, E, D+E).

3. Design and animals

Young adult male Wistar rats (about 8–10 weeks at surgery), from an ICAR-IVRI, NARFBR or other CPCSEA-registered facility (NIN Hyderabad no longer supplies animals; published rates are INR 300–500 per rat plus transport). Males only (no estrous variability; sex-as-variable limitation declared). Randomised by weight and source litter; coded syringes/vials; surgeon, injector and behaviour operator blind to drug; the environment arm cannot be fully blinded, so test-day scorers are separate from ELE handlers and scoring is automated where possible (partial blinding stated).

Arm	Drug package	Environment	Enrolled	Expected completers
C – peers	Vehicle (ICV + IP)	Standard (time-matched empty arena + partner)	16	~10–11
E	Vehicle	ELE	16	~10–11
D	FGF17 + 7,8-DHF	Standard	16	~10–11
D+E	FGF17 + 7,8-DHF	ELE	16	~10–11
Satellite	4 vehicle, 4 FGF17 bolus (engagement, 30 min later)	–	8	8
Partner rats	Unoperated stooge partners for arena sessions	–	10	–
Placement pilot	Cannula/dye practice and injection check	–	6	–
Total			88 rats	

Enrolment is 16 per arm because ICV-cannula attrition in rats is commonly 30–40% (cap loss, infection, exclusion for off-target placement); the plan assumes about 10–11 completers per arm (40–44). Exclusions are pre-specified: dye/histology-confirmed ventricular placement, >15% weight loss, humane endpoints, and a minimum of 3 of 4 tasks per animal. Housing: cannulated rats are pair-housed with a perforated divider if the ethics committee permits, otherwise single-housed, identically in all arms (the isolation confound is declared). Two staggered testing waves (32 rats each) are used because one operator cannot run four tasks on 64 rats at once; wave is a blocking factor.

Power (honest): with $N \approx 40$ –44 completers an interaction contrast reaches 80% power only for $f \approx 0.44$ –0.46, and a D+E vs C pairwise comparison only for $d \approx 1.3$ –1.4; power at $f = 0.25$ is only 0.34. Meta-analytic enrichment effects are small to moderate ($d \approx 0.3$ –0.45) and healthy young rats are near ceiling on simple tasks, so a null result is the likely outcome even if a modest true effect exists. The study is therefore interpreted by interval estimates, not significance tests, and pilot effect sizes must not be used to power the next experiment from the point estimate (Kraemer 2006; Leon 2011).

4. Interventions

Component	Specification
FGF17 (priming window, days 0–6)	Recombinant mouse FGF17 (alignment mouse vs rat to be done in silico before ordering; mouse FGF17 is cheaper and more widely listed than rat). Chronic guide cannula to the right lateral ventricle (scaled coordinates; dye + histology check). Four boluses on days 0, 2, 4, 6: 3 μ l of ~100 μ g/ml (0.3 μ g each; 1.2 μ g per rat) in aCSF + 0.1% BSA + heparin, delivered awake at 0.5–1 μ l/min with the injector left in place 2–3 min; aliquoted, thawed fresh each session, not left in the syringe at room temperature. Vehicle boluses in all other rats. <u>Deviation from the literature:</u> Iram 2022 used a continuous infusion for about a week; boluses give transient exposure, so "priming" is a hypothesis. Mandatory bench checks: ERK/proliferation bioassay of the supplied protein and the alignment. Upgrade path: Alzet 1007D/2004 pumps if budget or a discount allows (pumps are not otherwise affordable).
7,8-DHF	5 mg/kg IP daily, days 0–45 (through the whole test window), vehicle-matched. Rat precedent exists for 5 mg/kg (e.g. TBI studies), the cognition literature in rats is thin, and the mechanism is disputed. Solubility is checked on the bench first; $\leq 10\%$ DMSO with PEG300/Tween or a CMC-Na suspension (not 30% DMSO daily, which risks peritoneal irritation). Last dose 45 min before euthanasia in the western subset.
Enriched learning environment (ELE) — defined dose	Days 0–45, 1 h/day (after behaviour on test days) in a multi-level playground arena with an unoperated fixed partner rat, running wheels (exercise is part of the package and is a declared confound), tunnels, climbing structures, daily object rotation (≥ 3 new objects/day), and foraging/odour-cue and latch puzzle devices on a six-level difficulty ladder (promotion after 3 consecutive successes). Active engagement minutes and wheel revolutions are logged. <u>Transfer rule:</u> ELE puzzles are foraging/latch/odour devices only — no barrier or maze problems — so the Hebb-Williams transfer task is categorically different. Control arena: the same partner and handling, time-matched, without objects or puzzles.
BrdU (mechanism)	50 mg/kg IP on days 2, 4, 6 (100 mg/kg adds toxicity without a longer window); cheaper than EdU, but acid denaturation damages other epitopes, so BrdU is stained on adjacent sections from Olig2/MBP, with a single-slide pilot and positive control first.

5. Timeline (surgery day –7; day 0 = first FGF17/vehicle bolus)

Days	Activity
–7 to –1	Cannula surgery under anaesthesia with analgesia; recovery; dummy caps; habituation to handling and arena.
0–6	FGF17/vehicle boluses (days 0, 2, 4, 6); BrdU (days 2, 4, 6); ELE/control sessions and daily 7,8-DHF/vehicle begin on day 0; satellite engagement samples 30 min after a bolus.
14–21	Gradual food restriction to 85–90% of free-feeding weight (appetitive tasks only; body weight recorded as covariate).
22–29	Task 1: T-maze delayed alternation (working memory) — 5 days training, then delay testing.
30–36	Task 2: digging-based attentional set-shifting/reversal (flexibility) — habituation and training 3–5 days, test day(s).
37–40	Task 3: Hebb-Williams barrier maze battery (problem solving; the untrained transfer task).
41–45	Task 4: Morris water maze last (non-appetitive; 25 °C, rest warming; visible-platform control; swim speed and thigmotaxis). 4 acquisition days then probe on day 45.
46	Euthanasia (western subset 45 min after the final DHF dose); perfusion for the histology subset.

The fixed least-to-most-stressful task order is a declared limitation (carry-over cannot be separated from task). Two staggered waves offset by about two weeks make the in-life phase roughly 10 weeks. Treatment (ELE and DHF) continues through testing so drug, enrichment and testing windows coincide; FGF17 is the only agent with a finite priming window.

6. Endpoints

Aggregate performance index (primary, descriptive): mean of four task z-scores, using the control (C) arm mean and the pooled SD fixed a priori; signs aligned so that higher is better; animals need ≥ 3 of 4 tasks; body weight as covariate; wave as block. Task scores: delayed-alternation accuracy at long delays; set-shift trials-to-criterion (extradimensional shift); Hebb-Williams error score; MWM probe-trial target-quadrant time (with swim speed). The **inter-task correlation matrix is a primary descriptive output** — the real feasibility question is whether the tasks form a positive manifold at all; a factor analysis is out of scope at this n. NOR was dropped (ceiling in healthy young rats; one fewer task also limits fatigue).

Block	Readouts	n per arm
Behaviour (4 tasks)	See above; blinded automated scoring where possible.	all completers
Structure (perfusion subset)	BrdU+/Olig2+ cell density (dentate/CA1/corpus callosum; stereology or systematic sampling), MBP immunostaining; descriptive.	5
Engagement gates	Hippocampal pERK (and pTrkB only if an antibody is validated with a BDNF-stimulated control) 30–60 min after the last DHF dose; FGF17 engagement: pERK in the 8-rat satellite 30 min after a bolus.	4–5 (satellite 4 per group)

7. Statistics and pre-registration

- Register the plan on OSF before surgery; ARRIVE 2.0 reporting.
- Primary: linear model of the aggregate index with drug, environment and wave; planned contrasts D+E vs C, D+E vs E, D+E vs D (Holm); the interaction is exploratory. Effect sizes with 95% intervals.
- Missing-data rule: ≥ 3 of 4 tasks per animal; animals failing placement or humane criteria are excluded and reported (CONSORT-style flow).
- Repeated or nested data (acquisition, cage/housing) use mixed models; assay readouts are descriptive.

8. Pre-registered go/no-go rules for the larger experiment

Gate	Pass condition
Feasibility	$\geq 60\%$ of enrolled rats complete ≥ 3 of 4 tasks; cannula placement $\geq 75\%$ correct.
Reagent validity	FGF17 bioassay active; mouse/rat alignment documented; DHF solution stable at the dosing concentration.
FGF17 engagement	Satellite pERK at 30 min higher in FGF17 than vehicle (pre-set fold change, e.g. ≥ 1.5).
DHF/TrkB engagement	Validated pERK/pTrkB western shows increase vs vehicle; otherwise any TrkB-specific interpretation is withdrawn and the DHF component is reported as "effect of unestablished mechanism".
Direction	D+E – C index difference: point estimate $\geq +0.6$ SD, 95% CI lower bound > -0.3 SD, and D+E $\geq$ both E and D in point estimate.
Structure	BrdU+/Olig2+ density higher in drug arms (direction) and inter-task correlation matrix shows predominantly positive correlations.

If all gates pass, the larger experiment (Stage 1 in section 2) is justified: full factorial separating FGF17, 7,8-DHF and enrichment, both sexes, pumps, a sample size set from the pilot variance and a minimally important effect (for example, 0.5 SD needs about 64 rats per arm in a two-group comparison). If any gate fails, the failure mode (delivery, engagement, ceiling, attrition) decides the redesign; it is not evidence that the approach cannot work.

9. India logistics and budget

Item	What the review found
Ethics	Rats are "small animals": IAEC approval is sufficient (no central CPCSEA forwarding). IAEC meetings are often only 2/year to quarterly, proposals are due about 30 days ahead, and a missed meeting can cost months — ask for an interim review. Neonatal/lifespan work is not part of this protocol.
Animals	ICAR-IVRI Izatnagar INR 350; NISER INR 300–500 plus transport (certificates required). Retired breeders/aged rats are not routinely sold; not needed here.
Imports	FGF17 and antibodies are imported: expect 6–12 weeks, cold-chain risk, basic customs duty (about 10% for lab reagents) plus IGST (5–18%), unless the institute holds a DSIR certificate (Notification 51/96; reagent coverage not confirmed). Order early; a failed bioassay forces a re-order. The duty/GST figures are the reviewers' knowledge, not verified by search.
Timeline	About 10–14 months from IAEC submission to results: IAEC 1–6 months; procurement 2–3 (in parallel); pilots/bench checks 1–2; in-life ≈ 2.5 ; histology, westerns and analysis 2–3.

Budget (USD; estimates from list prices found by search and reviewer knowledge; the lab's existing maze, tracking, stereotaxic, microtome and microscope time are excluded; confirm every line with a quote):

Line	Estimate	Notes
Rats (88 incl. partners, satellite, practice) with transport	0.6k	INR 350–500 each + transport
Housing/care (about 60 days)	1.2k	per-diem not found; ask facility
FGF17: 3 × 25 µg (≈ 65 –75 µg incl. bioassay and waste) incl. duty, GST, cold chain	1.6k	list: 50 µg $\approx$ USD 480 (Cohesion); R&D; price not found; range 1.1–2.2k
7,8-DHF ≈ 3 g	0.4k	TCI $\approx$ USD 62/g; Sigma 100 mg $\approx$ USD 302
Guide cannulas, dummies, screws, dental cement	0.5k	in-house fabrication possible
Anaesthesia, analgesia, surgical and perfusion consumables	0.8k	
ELE arena, wheels, objects, puzzle devices	0.6k	
Behaviour consumables; Hebb-Williams and T-maze fabrication; pellets, digging media	0.6k	
BrdU	0.1k	USD 70–144 per 100–500 mg
Antibodies: BrdU, Olig2, MBP, pERK ($\pm$ pTrkB), secondaries	2.0k	USD 290–640 per vial list; validated pTrkB adds ≈ 0.5 k
Histology/western reagents, RNA-free consumables	0.7k	
IAEC fee and miscellaneous	0.1k	
Base total	9.2k	
Reserve	0.8k	$\approx 9\%$ — thin; see priority cuts
Planned envelope	≈ 10.0k	range 8.0–11.0k at current price uncertainty

Pumps are not affordable at this budget (Alzet 1007D $\approx$ USD 110–150 each, no Indian distributor found; 48 pumps $\approx$ USD 6k before duty), which is why FGF17 is given by cannula bolus. **Priority cuts if prices run high** (each preserves the behavioural question): drop the western block (≈ -1.1 k) and keep structure only; drop MBP staining (≈ -0.35 k); enrol 14 per arm instead of 16 (≈ -0.1 k, but attrition risk rises); drop pTrkB. **Must not be cut:** dye/histology placement check, FGF17 bioassay and alignment, blinding and randomisation, DHF solubility check, MWM visible-platform control, pre-registered go/no-go.

10. Validation record for this adaptation

Round	Draft	Verdicts	Main issues raised → fix
1	A (12-month retired breeders, Alzet 1007D pumps, 2×2 of FGF17 × DHF)	Design: approve with changes (as feasibility only). Pharmacology: approve with changes. Logistics: approve with changes.	Pump lasts 7 d, ends before testing; pumps cost ~USD 6k (cap breaker); 12-month rats are middle-aged, not aged; Golgi/western tissue conflict; NIN Hyderabad no longer supplies rats; aged retired breeders unobtainable; IAEC cadence → pumps replaced by cannula boluses, young adults used, Golgi dropped, new sourcing.
(user)	B/C (new aim: drug package × learning environment, composite across tasks)	–	Aim changed by the researcher; design rebuilt around enhancement in healthy young rats.
2	C	Design: approve with changes. Pharmacology: approve with changes. Enrichment/logistics: approve with changes.	"Transfer" task overlapped the enrichment puzzles → made categorically different; enrichment confounded with social contact and isolation → partner in both arenas, housing declared; composite not a g factor → renamed aggregate index, correlation matrix primary; attrition 30–40% → enrol 16/arm; timeline unrealistic (set-shifting, Hebb-Williams, MWM durations; DHF stopping mid-battery) → rebuilt, treatment through testing; NOR dropped; LFB and Golgi dropped; BrdU adjacent sections; satellite at 30 min; budget raised and cut list added.
(extra)	Translational goal section	Opus translational critic: approve with changes.	Overclaiming of "reconstruction" → "one possible contributor"; two separate programmes; neurotrophin delivery failures added; FGF23 point corrected to mitogenesis/angiogenesis; paediatric ethics and DSMB named; engagement and structural gates made explicit.

Only two review rounds were run for this adaptation (the earlier full protocol had five). Items still open are in section 11.

11. Items the reviewers still do not like / remaining gaps

#	Gap	Kind
1	The bolus-priming idea is biologically shaky (FGF17 is short-lived; Iram used continuous infusion). A null result may reflect delivery, not biology.	Cannot fix at this budget; pumps in the larger study.
2	Power: only very large effects are detectable ($d \approx 1.3$ – 1.4); realistic enrichment effects are $d \approx 0.3$ – 0.45 and healthy rats are near ceiling.	Intrinsic to a small pilot; interpret by intervals.
3	7,8-DHF may be pharmacologically inert in vivo (TrkB agonism disputed); the pTrkB antibody is notoriously unreliable.	Engagement gate; may withdraw TrkB claims.
4	The enrichment package confounds social contact, exercise and puzzles; single/divided housing is an isolation confound; environment arm cannot be fully blinded.	Declared; separated in the larger study.
5	The aggregate index is not a validated general factor; fixed task order; one operator; food restriction interacts with arm.	Declared; covariates and waves.
6	Prices not verified (R&D; FGF17, Alzet, per-diem, INR antibody prices, duty/GST); the budget could land at 8–11k; reserve is thin.	Needs quotes before IAEC submission.
7	The IAEC calendar (and a possible re-order of FGF17 if the bioassay fails) can add months.	Schedule risk.
8	No molecular evidence of "neuroplasticity" (no LTP/spines); structure is descriptive only at $n \approx 5$.	Stage 1.
9	Literature checks were from search snippets only; several references are partial (section 12). The human-translation claims depend on facts to verify (no human trials of FGF17/7,8-DHF found; neurotrophin delivery failures).	Researcher to verify before submission.

12. References and verification status

All checks were by search snippets (PubMed, PMC, Crossref and publisher pages were unreachable). "Verified" means consistent across indexed records, not checked against the publisher page.

#	Citation	Status
1	Iram T, Kern F, Kaur A, et al. Young CSF restores oligodendrogenesis and memory in aged mice via Fgf17. <i>Nature</i> 2022;605:509–515. doi:10.1038/s41586-022-04722-0. (Fgf17 infused ~1 week in aged mice; dose and pump details not retrieved.)	Verified; methods not accessed
2	Boltaev U, et al. Multiplex quantitative assays indicate a need for reevaluating reported small-molecule TrkB agonists. <i>Sci Signal</i> 2017;10(493). doi:10.1126/scisignal.aal1670	Verified (DOI from journal URL)
3	Todd D, Gowers I, Dowler SJ, et al. A monoclonal antibody TrkB receptor agonist as a potential therapeutic for Huntington's disease. <i>PLoS One</i> 2014;9(2):e87923. doi:10.1371/journal.pone.0087923	Citation verified; 7,8-DHF content per secondary reviews
4	Pharmaceuticals 2021;14(8):704 — review of whether small molecules activate TrkB like BDNF.	Verified (review)
5	Rawlings-Mortimer F, et al. 7,8-DHF during water-maze training in wild-type mice. <i>Front Syst Neurosci</i> 2023. doi:10.3389/fnsys.2023.1134594	Partial (authors from snippet)
6	van Praag H, Kempermann G, Gage FH. Neural consequences of environmental enrichment. <i>Nat Rev Neurosci</i> 2000;1:191–198. doi:10.1038/35044558	Verified
7	Nithianantharajah J, Hannan AJ. Enriched environments, experience-dependent plasticity and disorders of the nervous system. <i>Nat Rev Neurosci</i> 2006;7:697–709. doi:10.1038/nrn1970	Verified
8	Galsworthy MJ, et al. Assessing reliability, heritability and general cognitive ability in a battery of cognitive tasks for laboratory mice. <i>Behav Genet</i> 2005;35:675–692. doi:10.1007/s10519-005-3423-9	Verified
9	Matzel LD, Han YR, Grossman H, et al. Individual differences in the expression of a "general" learning ability in mice. <i>J Neurosci</i> 2003.	Partial (volume/pages not verified)
10	Birrell JM, Brown VJ. Medial frontal cortex mediates perceptual attentional set shifting in the rat. <i>J Neurosci</i> 2000;20(11):4320–4324.	Verified
11	Rabinovitch MS, Rosvold HE. A closed field intelligence test for rats. <i>Can J Psychol</i> 1951;5(3):122–128. (Hebb-Williams maze.)	Partial; no modern validation found
12	Kraemer HC, et al. Caution regarding the use of pilot studies to guide power calculations. <i>Arch Gen Psychiatry</i> 2006;63:484–489. doi:10.1001/archpsyc.63.5.484	Verified
13	Leon AC, Davis LL, Kraemer HC. The role and interpretation of pilot studies in clinical research. <i>J Psychiatr Res</i> 2011;45:626–629. doi:10.1016/j.jpsychires.2010.10.008	Verified
14	Percie du Sert N, et al. The ARRIVE guidelines 2.0. <i>PLoS Biol</i> 2020;18:e3000410. doi:10.1371/journal.pbio.3000410	Verified
15	Léger M, et al. 2014, duration of enrichment effects in mice (effects fade within weeks after stopping); and an enrichment meta-analysis in <i>Biol Rev</i> 96(6).	Partial (citations not fully retrieved)

Not found: any study of FGF17/FGF-family treatment improving learning in healthy young adult rodents; any study combining 7,8-DHF with environmental enrichment; rat ICV FGF17 bolus doses; an Indian distributor and INR prices for Alzet pumps and recombinant FGF17; a peer-reviewed modern validation of the Hebb-Williams maze; per-diem housing cost.